

Rohit Kumar

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LinkedIn | Portfolio | GitHub

Professional Summary

AI Application Developer with experience building agentic AI workflows, RAG pipelines, cloud-native microservices, and enterprise automation. Hands-on with Python, Java, FastAPI, LangChain, LangGraph, Kubernetes, OCI, OKE, KServe, Kubeflow Pipelines, vector databases, and LLM-based systems. Google Summer of Code 2026 contributor for Kubeflow, focused on Agentic RAG, MCP tools, ingestion pipelines, and reproducible deployment on OCI.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

AI, LLMs, and Agents: LangChain, LangGraph, RAG, MCP, Kagent, Prompt Engineering, Llama, Autogen, PandasAI, QLoRA/LoRA

Backend and APIs: FastAPI, Node.js, Express.js, REST APIs, Java ADF, Spring Boot basics

Cloud, DevOps, and MLOps: OCI, OKE, Kubernetes, Kubeflow Pipelines, KServe, Docker, GitHub Actions, CI/CD, Git, Postman

Databases and Search: PostgreSQL, MySQL, Elasticsearch, Milvus, FAISS, ChromaDB, Firebase, Supabase, Prisma ORM

Frontend: React, Next.js, TypeScript, Tailwind CSS, ReactFlow, ShadCN UI

Professional Experience

Oracle - AI Application Developer, Oracle Applications Labs (OAL)

Apr 2026 – Present

Hyderabad, India

- Added web-search enhancement for MPN matching when fuzzy matching and clustering did not produce sufficient matches.
- Integrated catalog lookup across official and reliable third-party retailer sources to improve asset and part matching coverage for ACT workflows.
- Supported downstream reconciliation and migration-readiness analysis by improving discovery of manufacturer part information.
- Built a LangGraph proof of concept to evaluate moving the ACT agent from Codex-style skill execution to multi-agent orchestration.
- Converted Codex skill concepts into specialized worker-agent responsibilities and evaluated routing, maintainability, and LangChain/LangGraph tradeoffs.

Google Summer of Code 2026 - Kubeflow - Open Source Contributor, Agentic RAG on Kubeflow

May 2026 –

Present

Remote

- Accepted for Kubeflow GSoC 2026 project to expand `kubeflow/docs-agent` into an Agentic RAG reference architecture.
- Building multi-index retrieval across documentation, GitHub issues, and Kubeflow manifests using Python, MCP tools, Kagent, Milvus, and vector search.
- Developed ingestion pipelines with Kubeflow Pipelines and KServe deployment patterns; live-tested docs-agent stack on OCI/OKE.
- Indexed 1,631 GitHub issue chunks and 477 manifest/code chunks, improving semantic search coverage for technical Kubeflow questions.
- Created a 71-test suite and CI workflow foundation to improve reliability, parser coverage, and maintainability for the open-source project.

Oracle - Application Developer, Supply Chain Management (SCM)

Jun 2024 – Apr 2026

Hyderabad, India

- Designed AI agent workflows for Oracle Fusion Supply Chain Planning using Python, LangChain, PandasAI, and enterprise data sources.
- Built multi-agent query-resolution flows for plan exceptions, note generation, and pivot-table analysis on planning data.
- Implemented order recommendation workflow to classify orders for release or review using business rules, exceptions, notes, and batch ESS processing.
- Led a RAG chatbot with Llama 3.1 and vector search to provide real-time lead-time insights from supply chain datasets.
- Contributed Java ADF and SQLDeveloper features for Supply Chain Planning and supported Gen AI Hackathon-winning enterprise prototype.

Oracle - Application Developer Intern

May 2023 – Jul 2023

Hyderabad, India

- Developed copy-data functionality for lead-time columns and optimized retrieval using Elasticsearch and PATCH updates.
- Added linear-regression trend analysis in Oracle JET charts for supply chain visualization.

Projects

FlowForge - Visual Agent Workflow Builder

FastAPI, React, ReactFlow, ChromaDB, FAISS, Supabase

- Built FastAPI microservices and RAG pipelines using ChromaDB, FAISS, and Together AI for LLM inference.
- Developed React/ReactFlow UI with Supabase PostgreSQL and Prisma for visual agent workflow configuration.

ChemTutor LLM - Fine-tuned Chemistry Language Model

Llama 7B, QLoRA, Hugging Face, PyTorch

- Fine-tuned Llama 7B on Class 12 Chemistry NCERT content using QLoRA 4-bit quantization on a 4GB VRAM GPU.
- Published adapters to Hugging Face, enabling integration with compatible base LLMs for student doubt-solving use cases.

Code It NITW - Programming Learning Platform

React, Tailwind CSS, Firebase, Chart.js

- Built platform used by 200+ daily users for contests, interview experiences, and structured DSA preparation.
- Integrated Codeforces API, Firebase authentication, and Chart.js rankings for real-time competitive programming insights.

Education

National Institute of Technology, Warangal - B.Tech in Mechanical Engineering
CGPA: 7.34

2020 - 2024

Achievements and Leadership

- Won Oracle Gen AI Hackathon: 1st among 33 teams and top 5 among 300+ employees.
- Runner-Up, Tri-NIT Hackathon 2024 in Backend Development track.
- Solved 600+ programming problems across LeetCode, GeeksforGeeks, CodeChef, and Codeforces.
- Placement Trainer, Training and Placement Cell, NIT Warangal; Executive Member, CodeChef Chapter NITW.